

Exp.No.7-Incident Timeline Reconstruction Using System Logs in Kali Linux

Aim

To reconstruct the sequence of events during a security incident by analyzing system logs and arranging activities in chronological order.

Objective

- To understand the importance of timeline reconstruction
- To extract event timestamps from system logs
- To identify suspicious activities
- To arrange events chronologically for incident investigation

```
(unknown@kali):~$ sudo journalctl | grep login
Jul 17 17:40:43 kali systemd[1]: Listening on systemd-logind-varlink.socket - User Login Management Varlink Socket.
Jul 17 17:40:43 kali systemd[1]: getty-static.service - getty on tty2-tty6 if dbus and logind are not available skipped, umet condition check ConditionPathExists=!/usr/bin/dbus-daemon
Jul 17 17:40:44 kali systemd[1]: Starting systemd-logind.service - User Login Management...
Jul 17 17:40:44 kali systemd-logind[538]: New seat seat0.
Jul 17 17:40:44 kali systemd-logind[538]: Watching system buttons on /dev/input/event1 (Power Button)
Jul 17 17:40:44 kali systemd-logind[538]: Watching system buttons on /dev/input/event2 (Sleep Button)
Jul 17 17:40:44 kali systemd-logind[538]: Watching system buttons on /dev/input/event0 (AT Translated Set 2 keyboard)
Jul 17 17:40:44 kali systemd[1]: Started systemd-logind.service - User Login Management.
Jul 17 17:40:45 kali systemd-logind[538]: New session 'c1' of user 'lightdm' with class 'greeter' and type 'x11'.
Jul 17 17:40:45 kali systemd-logind[538]: New session '1' of user 'lightdm' with class 'manager-early' and type 'unspecified'.
Jul 17 17:40:46 kali rtkit-daemon[765]: Handling system-suspend using logind.
Jul 17 17:40:57 kali systemd-logind[538]: Removed session c1.
Jul 17 17:48:57 kali systemd-logind[538]: New session '3' of user 'unknown' with class 'user' and type 'x11'.
Jul 17 17:48:57 kali systemd-logind[538]: New session '4' of user 'unknown' with class 'manager' and type 'unspecified'.
Jul 17 17:48:57 kali lightdm[969]: gkr-pam: unlocked login keyring
Jul 17 17:49:07 kali systemd-logind[538]: Removed session 1.
Jul 17 17:55:54 kali systemd-logind[538]: The system will power off now!
Jul 17 17:55:54 kali systemd-logind[538]: System is powering down.
Jul 17 17:55:54 kali systemd-logind[538]: Removed session 3.
Jul 17 17:55:54 kali systemd-logind[538]: Removed session 4.
Jul 17 17:55:55 kali systemd[1]: Stopping systemd-logind.service - User Login Management...
Jul 17 17:55:55 kali systemd[1]: systemd-logind.service: Deactivated successfully.
Jul 17 17:55:55 kali systemd[1]: Stopped systemd-logind.service - User Login Management.
Jul 17 17:55:55 kali systemd[1]: systemd-logind-varlink.socket: Deactivated successfully.
Jul 17 17:55:55 kali systemd[1]: Closed systemd-logind-varlink.socket - User Login Management Varlink Socket.
Jul 17 20:47:45 kali systemd[1]: Listening on systemd-logind-varlink.socket - User Login Management Varlink Socket.
Jul 17 20:47:45 kali systemd[1]: getty-static.service - getty on tty2-tty6 if dbus and logind are not available skipped, umet condition check ConditionPathExists=!/usr/bin/dbus-daemon
Jul 17 20:47:46 kali systemd[1]: Starting systemd-logind.service - User Login Management...
Jul 17 20:47:46 kali systemd-logind[493]: New seat seat0.
Jul 17 20:47:46 kali systemd-logind[493]: Watching system buttons on /dev/input/event1 (Power Button)
Jul 17 20:47:46 kali systemd-logind[493]: Watching system buttons on /dev/input/event2 (Sleep Button)
Jul 17 20:47:46 kali systemd-logind[493]: Watching system buttons on /dev/input/event0 (AT Translated Set 2 keyboard)
Jul 17 20:47:46 kali systemd[1]: Started systemd-logind.service - User Login Management.
Jul 17 20:47:47 kali systemd-logind[493]: New session 'c1' of user 'lightdm' with class 'greeter' and type 'x11'.
Jul 17 20:47:48 kali systemd-logind[493]: New session '1' of user 'lightdm' with class 'manager-early' and type 'unspecified'.
Jul 17 20:47:48 kali rtkit-daemon[731]: Handling system-suspend using logind.
Jul 17 20:59:38 kali systemd-logind[493]: Removed session c1.
Jul 17 20:59:38 kali systemd-logind[493]: New session '3' of user 'unknown' with class 'user' and type 'x11'.
Jul 17 20:59:38 kali systemd-logind[493]: New session '4' of user 'unknown' with class 'manager' and type 'unspecified'.
Jul 17 20:59:39 kali lightdm[956]: gkr-pam: unlocked login keyring
Jul 17 20:59:48 kali systemd-logind[493]: Removed session 1.
Jul 17 21:01:01 kali systemd-logind[493]: The system will power off now!
Jul 17 21:01:01 kali systemd-logind[493]: System is powering down.
Jul 17 21:01:02 kali systemd-logind[493]: Session 3 logged out. Waiting for processes to exit.
Jul 17 21:01:02 kali systemd-logind[493]: Removed session 3.
Jul 17 21:01:02 kali systemd-logind[493]: Removed session 4.
```

```

Aug 22 13:33:05 kali sudo[10438]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:33:05 kali sudo[10438]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl --since -10m

(unknown@kali)-[~]
$ sudo journalctl --since "-10m"
Aug 22 13:25:01 kali CRON[6428]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Aug 22 13:25:01 kali CRON[6430]: (root) CMD (command -v debian-sal1 > /dev/null && debian-sal1 1)
Aug 22 13:25:01 kali CRON[6428]: pam_unix(cron:session): session closed for user root
Aug 22 13:25:08 kali sudo[6450]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl -xe
Aug 22 13:25:08 kali sudo[6450]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:25:58 kali sudo[6450]: pam_unix(sudo:session): session closed for user root
Aug 22 13:26:59 kali sudo[7424]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl
Aug 22 13:26:59 kali sudo[7424]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:27:00 kali sudo[7424]: pam_unix(sudo:session): session closed for user root
Aug 22 13:27:13 kali sudo[7546]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl
Aug 22 13:27:13 kali sudo[7546]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:27:14 kali sudo[7546]: pam_unix(sudo:session): session closed for user root
Aug 22 13:27:36 kali systemd-timesyncd[423]: Timed out waiting for reply from 3.6.43.90:123 (2.debian.pool.ntp.org).
Aug 22 13:27:47 kali systemd-timesyncd[423]: Timed out waiting for reply from 17.253.18.131:123 (2.debian.pool.ntp.org).
Aug 22 13:27:57 kali systemd-timesyncd[423]: Timed out waiting for reply from [2600:3c08::2000:9bff:fe44:8ddb]:123 (2.debian.pool.ntp.org).
Aug 22 13:28:07 kali systemd-timesyncd[423]: Timed out waiting for reply from [2606:4700:f1::123]:123 (2.debian.pool.ntp.org).
Aug 22 13:28:17 kali systemd-timesyncd[423]: Timed out waiting for reply from [2a01:4f9:c010:c22::1]:123 (2.debian.pool.ntp.org).
Aug 22 13:28:19 kali sudo[8111]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl --since 10m
Aug 22 13:28:19 kali sudo[8111]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:28:19 kali sudo[8111]: pam_unix(sudo:session): session closed for user root
Aug 22 13:28:28 kali systemd-timesyncd[423]: Timed out waiting for reply from [2605:e440:19::1e7]:123 (2.debian.pool.ntp.org).
Aug 22 13:29:18 kali sudo[8603]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl --since 10m
Aug 22 13:29:18 kali sudo[8603]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:29:18 kali sudo[8603]: pam_unix(sudo:session): session closed for user root
Aug 22 13:31:32 kali systemd[1]: Starting systemd-tmpfiles-clean.service - Cleanup of Temporary Directories ...
Aug 22 13:31:33 kali systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Aug 22 13:31:33 kali systemd[1]: Finished systemd-tmpfiles-clean.service - Cleanup of Temporary Directories.
Aug 22 13:31:48 kali systemd-timesyncd[423]: Contacted time server 95.216.144.226:123 (0.debian.pool.ntp.org).
Aug 22 13:32:48 kali sudo[10289]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl --since -10m
Aug 22 13:32:48 kali sudo[10289]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:32:48 kali sudo[10289]: pam_unix(sudo:session): session closed for user root
Aug 22 13:33:05 kali sudo[10438]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl --since -10m
Aug 22 13:33:05 kali sudo[10438]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:33:05 kali sudo[10438]: pam_unix(sudo:session): session closed for user root
Aug 22 13:34:06 kali sudo[10945]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl --since -10m
Aug 22 13:34:06 kali sudo[10945]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)

```

```

Session Actions Edit View Help
(unknown@kali)-[~]
$ date
Saturday 22 August 2026 01:21:21 PM IST

(unknown@kali)-[~]
$ whoami
unknown

(unknown@kali)-[~]
$ su root
Password:
su: Authentication failure

(unknown@kali)-[~]
$ sudo journalctl -xe
[sudo] password for unknown:
Aug 22 13:17:09 kali systemd[938]: Starting xdg-desktop-portal-gtk.service - Portal service (GTK/GNOME implementation) ...
Subject: A start job for unit UNIT has begun execution
Defined-By: systemd
Support: https://www.debian.org/support

A start job for unit UNIT has begun execution.

The job identifier is 538.
Aug 22 13:17:09 kali dbus-daemon[963]: [session uid=1000 pid=963 pidfd=5] Successfully activated service 'org.freedesktop.impl.portal.desktop.gtk'
Aug 22 13:17:09 kali systemd[938]: Started xdg-desktop-portal-gtk.service - Portal service (GTK/GNOME implementation).
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: https://www.debian.org/support

A start job for unit UNIT has finished successfully.

The job identifier is 538.
Aug 22 13:17:09 kali dbus-daemon[963]: [session uid=1000 pid=963 pidfd=5] Successfully activated service 'org.freedesktop.portal.Desktop'
Aug 22 13:17:09 kali systemd[938]: Started xdg-desktop-portal.service - Portal service.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: https://www.debian.org/support

A start job for unit UNIT has finished successfully.

The job identifier is 460.
Aug 22 13:17:10 kali rtkit-daemon[784]: Successfully made thread 2117 of process 1902 owned by '1000' RT at priority 10.
Aug 22 13:17:39 kali systemd[1]: systemd-timedated.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://www.debian.org/support

The unit systemd-timedated.service has successfully entered the 'dead' state.
Aug 22 13:19:16 kali unix_chkpwd[3376]: password check failed for user (root)
Aug 22 13:19:16 kali su[3358]: pam_unix(su:auth): authentication failure; logname=unknown uid=1000 euid=0 tty=/dev/pts/0 ruser=unknown rhost= user=root
Aug 22 13:19:18 kali su[3358]: FAILED SU (to root) unknown on pts/0
Aug 22 13:19:51 kali sudo[3592]:  unknown : TTY=pts/0 ; PWD=/home/unknown ; USER=root ; COMMAND=/usr/bin/journalctl -xe
Aug 22 13:19:51 kali sudo[3592]: pam_unix(sudo:session): session opened for user root(uid=0) by unknownm(uid=1000)
Aug 22 13:20:24 kali kernel: 07:50:24.926370 SHCLX11 Shared Clipboard: Converting VBox Formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Aug 22 13:20:24 kali kernel: 07:50:24.930966 SHCLX11 Shared Clipboard: Converting VBox Formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
Aug 22 13:20:47 kali systemd-timesyncd[423]: Timed out waiting for reply from 157.245.102.2:123 (2.debian.pool.ntp.org).
Aug 22 13:20:58 kali systemd-timesyncd[423]: Timed out waiting for reply from 192.46.211.253:123 (2.debian.pool.ntp.org).
Aug 22 13:20:58 kali systemd-timesyncd[423]: Contacted time server 3.6.43.90:123 (2.debian.pool.ntp.org).
Aug 22 13:21:11 kali sudo[3592]: pam_unix(sudo:session): session closed for user root
Aug 22 13:21:40 kali unix_chkpwd[4814]: password check failed for user

```